

SOFTWARE REQUIREMENTS SPECIFICATION

Leave Management System

IEEE-Style SRS Document

Prepared from the project documentation and elicitation document

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the Leave Management System. The system is intended to provide a centralized web-based platform for employees to submit leave requests, for managers or designated approvers to review and approve or reject requests, and for administrators to maintain leave policies, balances, holidays and organizational leave records.

1.2 Scope

The Leave Management System is a web-based application designed to digitize and automate the leave request and approval workflow within a single organization. Employees can apply for leave by selecting a leave type, date range and reason, with supporting document upload where applicable. The system routes requests to the appropriate approving authority, records approval or rejection decisions, updates leave balances and history, and provides notifications and calendar-based visibility.

The system also provides an administrative dashboard for managing leave policies, public holidays and department settings, together with basic reports and analytics. It is intended to reduce delays, manual errors and administrative workload while improving transparency of employee leave information.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
LMS	Leave Management System
SRS	Software Requirements Specification
HR	Human Resources
RBAC	Role-Based Access Control
Leave Balance	Remaining leave entitlement available to an employee
Approver	Manager or authorized person responsible for reviewing leave requests
Leave History	Recorded history of submitted, approved, rejected or otherwise processed leave requests

1.4 References

- Leave Management System – Project Documentation
- Leave Management System – Elicitation Document

The requirements in this SRS are derived from the supplied project documentation and the corresponding elicitation material.

2. Overall Description

2.1 Product Perspective

The proposed system is a standalone web-based Leave Management System for a single organization instance. It consists of a responsive user interface, authentication and role-management module, employee leave application module, approval workflow module, leave balance and history module, notification component, calendar view, and an administrative dashboard backed by a relational database.

2.2 Product Functions

- Allow employees to register, authenticate and access the system according to their assigned role.
- Allow employees to submit leave applications with leave type, date range, reason and supporting document where applicable.
- Route leave requests to the designated approving authority.
- Allow managers or approvers to review, approve or reject leave requests.

- Calculate and maintain employee leave balances, accruals and leave history.
- Provide email or in-app notifications for leave application status updates.
- Provide a calendar view of approved leaves for teams and departments.
- Allow administrators to configure leave policies, public holidays and department settings.
- Provide basic leave reports and department-wise analytics.
- Protect employee information through authentication and role-based access control.

2.3 User Classes and Characteristics

User Class	Role / Needs
Employee	Primary user; applies for leave, checks leave balance and history, and tracks request status.
Manager / Approver	Reviews team leave requests and approves or rejects them according to organizational policy.
Administrator / HR	Maintains leave policies, holidays, department settings, employee records relevant to leave processing and reports.

2.4 Operating Environment

- Responsive web application accessible through standard desktop and mobile web browsers.
- A server-side application/API layer for authentication, leave processing and workflow management.
- A relational database for employee, leave, policy, holiday and approval records.
- An email or in-app notification mechanism for workflow status updates.
- The system can be developed and tested on standard personal computers/laptops without specialized hardware.

2.5 Design and Implementation Constraints

- The system is designed for a single organization instance.
- Access to leave information and administrative functions shall be controlled by role-based permissions.
- Leave balances and application status shall be maintained as centralized digital records.
- The delivered solution is a responsive web application; native mobile applications are outside the current scope.
- Third-party payroll/salary systems, biometric attendance hardware and external HRMS integrations are outside the current scope.
- Advanced AI-based leave prediction or workforce optimization is outside the current scope.

2.6 Assumptions and Dependencies

- Employees, managers and administrators have access to a supported browser and valid account credentials.
- The organization provides applicable leave types, policies, holidays and approval hierarchy.
- Authorized administrators are available to maintain current leave policies and organizational settings.
- A relational database and standard web application hosting environment are available.
- Email or in-app notification infrastructure is available where notifications are enabled.

3. Functional Requirements

3.1 Authentication and Role Management

FR-01: The system shall provide secure user authentication for employees, managers/approvers and administrators.

FR-02: The system shall enforce role-based access so that users can access only functions permitted for their role.

3.2 Leave Application Submission

- FR-03: The system shall allow employees to submit leave applications online.
- FR-04: The system shall capture leave type, start date, end date and reason for each leave request.
- FR-05: The system shall allow supporting documents to be uploaded where required by the leave policy.
- FR-06: The system shall validate leave application information before submission.

3.3 Leave Approval and Rejection

- FR-07: The system shall route submitted leave requests to the appropriate approving authority.
- FR-08: The system shall allow authorized managers/approvers to review pending leave requests.
- FR-09: The system shall allow authorized approvers to approve or reject leave requests.
- FR-10: The system shall record the decision and updated request status.

3.4 Leave Balance and History

- FR-11: The system shall calculate and display the current leave balance for each employee.
- FR-12: The system shall maintain leave accrual and leave usage records as configured by policy.
- FR-13: The system shall provide employees with their leave history and request status.

3.5 Notifications

- FR-14: The system shall notify employees when a leave request is submitted, approved or rejected.
- FR-15: The system shall notify relevant approvers when a new leave request requires action.

3.6 Calendar and Scheduling

- FR-16: The system shall provide a calendar view of approved leaves for teams and departments.
- FR-17: The system shall allow authorized users to view relevant leave schedules for workforce planning.

3.7 Administration and Reporting

- FR-18: The system shall allow administrators to configure leave types and leave policies.
- FR-19: The system shall allow administrators to manage public holidays and department settings.
- FR-20: The system shall provide basic reports and analytics such as leave trends and department-wise summaries.

4. Non-Functional Requirements

4.1 Performance

- NFR-01: The system should respond quickly enough to support practical day-to-day leave application and approval operations.
- NFR-02: Common actions such as login, leave submission and viewing balances should complete without unnecessary delay.

4.2 Usability

- NFR-03: The interface should be simple and understandable for employees, managers and administrators.
- NFR-04: The leave application and approval workflows should require minimal training.

4.3 Availability and Accessibility

- NFR-05: The web application should be accessible during normal organizational operations and from supported desktop and mobile browsers.

4.4 Maintainability

NFR-06: Leave policies, holiday calendars and department settings should be maintainable without redesigning the application.

NFR-07: The system should use a modular architecture so additional leave types or workflow changes can be introduced with minimal disruption.

4.5 Security and Confidentiality

NFR-08: Employee leave information shall be accessible only to authenticated and authorized users.

NFR-09: Administrative configuration functions shall be restricted to authorized administrators.

NFR-10: Access to data shall follow role-based permissions.

4.6 Data Integrity and Accuracy

NFR-11: The system shall maintain accurate and consistent leave balances, request statuses and history records.

NFR-12: Approval and rejection actions shall be recorded against the corresponding leave request.

4.7 Reporting and Auditability

NFR-13: Leave records shall be centrally maintained to support reporting, auditing and organizational planning.

4.8 Scope Compliance

NFR-14: The system shall support leave management activities and shall not include payroll integration, biometric hardware integration, external HRMS integration, native mobile applications or advanced AI-based leave prediction within the current project scope.

5. External Interface Requirements

5.1 User Interface

- A responsive web interface for employees, managers/approvers and administrators.
- Employee screens for login, dashboard, leave application, leave balance, leave history and request status.
- Manager/approver screens for pending requests, request details and approval/rejection actions.
- Administrative screens for leave policies, holidays, departments and reports.
- A calendar view for approved team and department leaves.

5.2 Software Interfaces

- Relational database for employee, leave, policy, approval, holiday and reporting data.
- Server-side application/API layer for authentication, business logic and workflow processing.
- Email and/or in-app notification service for status updates and approval alerts, where enabled.

5.3 Database Interface

The system shall store and retrieve employee details relevant to leave processing, leave types, policies, applications, approvals, leave balances, leave history, public holidays, departments and notification status. The database shall maintain referential integrity between related leave records.

5.4 Communication Interface

The system is intended to operate through standard web communication between the client browser and application server. Specific network protocols, API endpoints and deployment security configurations are not defined in the supplied project documentation and therefore remain design-level decisions.

6. Other Requirements

6.1 Information and Content Requirements

- The system shall maintain current leave types and organizational leave policies.
- The system shall maintain public holiday information used for leave processing or calendar display.
- The system shall maintain employee leave balances and history for supported leave categories.
- Reports should provide basic summaries such as leave trends and department-wise information.

6.2 Authentication and Authorization

- Authentication shall be required for access to protected leave information and actions.
- Only authorized managers/approvers may process leave requests assigned to them.
- Only authorized administrators may change leave policies, holidays and department settings.

6.3 Out-of-Scope Requirements

- Integration with third-party payroll or salary-processing systems.
- Biometric or physical attendance tracking hardware integration.
- Multi-organization / multi-tenant support.
- Native mobile applications.
- Advanced AI-based leave prediction or workforce optimization.
- Integration with external HRMS platforms.

7. Appendices

Appendix A – Acceptance Criteria

ID	Requirement	Acceptance Condition
AC-01	User Authentication	An employee, manager/approver or administrator can log in and access the functions permitted for their role.
AC-02	Leave Application	An employee can submit a leave request with leave type, date range and reason.
AC-03	Supporting Document	The system accepts a supporting document for leave types where documentation is required.
AC-04	Approval Workflow	A submitted leave request is routed to the designated approving authority and can be approved or rejected.
AC-05	Leave Balance	The system displays the employee's current leave balance and maintains it after processed leave requests.
AC-06	Leave History	An employee can view submitted requests and their current or historical status.
AC-07	Notifications	Relevant users receive an email or in-app status notification when defined workflow events occur.
AC-08	Calendar	Authorized users can view approved leaves in a team/department calendar.
AC-09	Administration	An authorized administrator can configure leave policies, public holidays and department settings.
AC-10	Reporting	An authorized administrator can generate basic leave summaries or department-wise reports.

AC-11	Security	Protected leave information and administrative features are available only to authenticated and authorized users.
AC-12	Scope Compliance	The system does not include payroll integration, biometric hardware integration, multi-tenant support, native mobile apps, advanced AI leave prediction or external HRMS integration within this project.

Appendix B – Key Stakeholders

Stakeholder	Primary Interest
Employees	Simple online leave application, accurate leave balances, transparent status tracking and accessible leave history.
Managers / Approvers	Efficient review and approval/rejection of team leave requests and visibility into team schedules.
HR / Administrators	Centralized policy management, accurate leave records, holiday configuration and reporting.
Organization / Management	Reduced manual workload, improved transparency and better leave/workforce planning.

Appendix C – Key Findings from Elicitation

- A centralized digital leave process is needed to reduce delays and manual handling associated with paper-based or spreadsheet-based processing.
- Employees need direct visibility into leave balances, request status and leave history.
- Managers/approvers need a structured workflow for reviewing and deciding on leave requests.
- HR and administrators need configurable leave policies, holiday settings and centralized reporting.
- Automated notifications can reduce the need for repeated status queries and manual follow-up.
- A calendar view can support team and department-level leave visibility and workforce planning.
- Role-based access is necessary to protect employee information and restrict administrative actions.
- The system is intended to digitize and streamline leave management rather than integrate with payroll, biometric attendance or external HRMS platforms.

Appendix D – Expected Outcome

- A functional web-based platform that digitizes the leave application and approval lifecycle.
- Reduced processing time and administrative effort compared with manual leave handling.
- Improved accuracy and transparency in leave balance and leave history tracking.
- A centralized repository of leave data supporting reporting, auditing and organizational planning.
- A maintainable architecture that can support future additions such as new leave types or approval workflows.